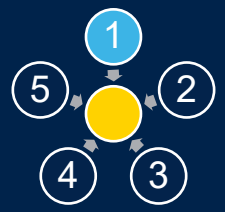
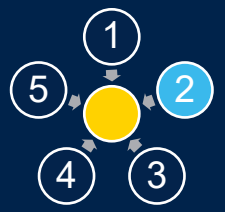
Select the right part and start your POC

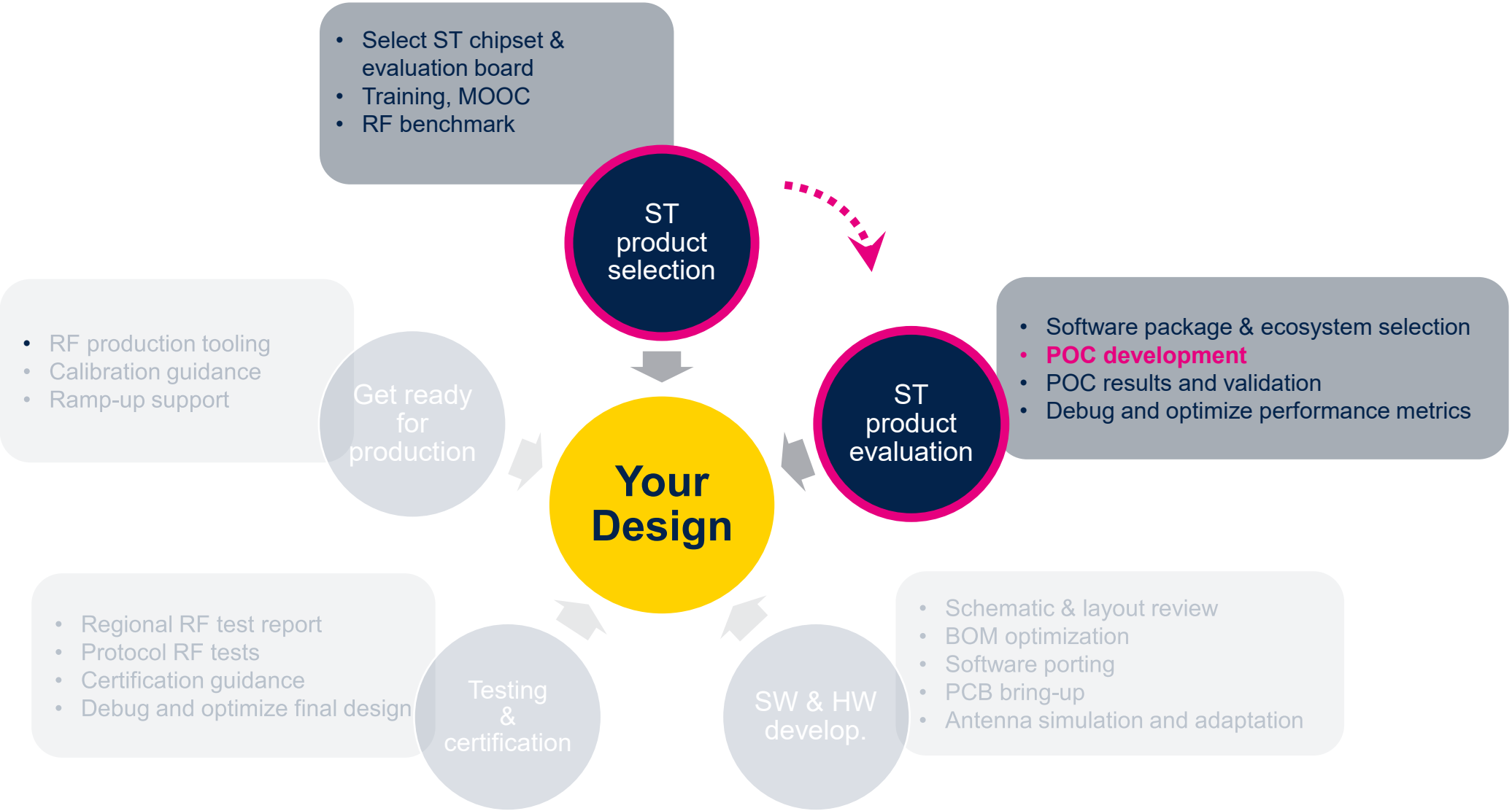


ST Product Selection

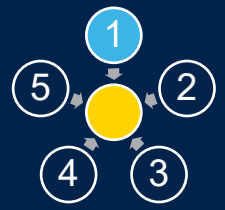


ST Product Evaluation

ST support your **wireless journey**
at each project development phase



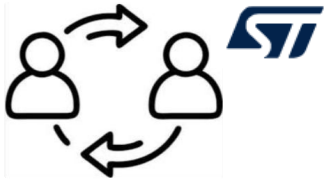
ST Product Selection



ST Product Selection

Understand application needs

You



Thanks to technical exchanges understand customer application, translate needs into technical specifications, and select the right wireless technology

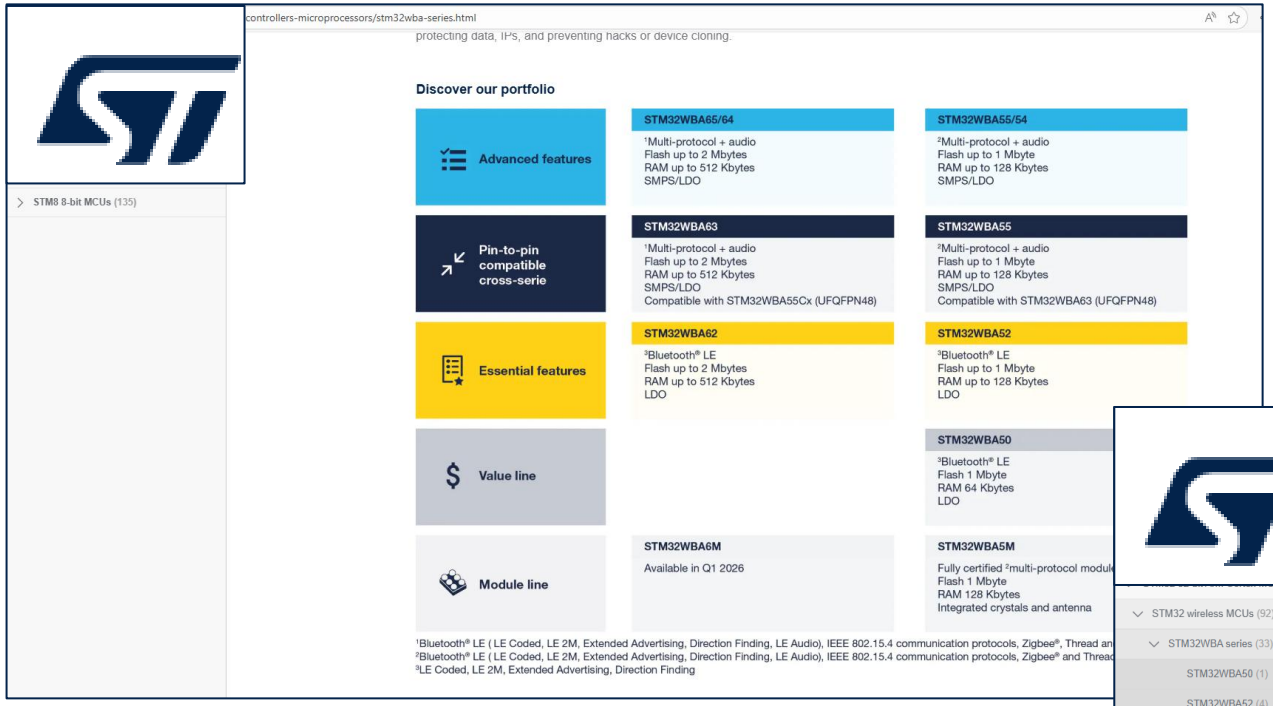
With our support select the appropriate part number based on application constraints, power efficiency, bill of materials, price, and size constraints.

Select the right part (1/2)

“We are designing sensor nodes and plan to enhance our solution to Bluetooth® Low Energy. The application has strict power constraints and requires flash memory to store data. Which device is recommended, and how can you help to evaluate and demonstrate capabilities ? ”

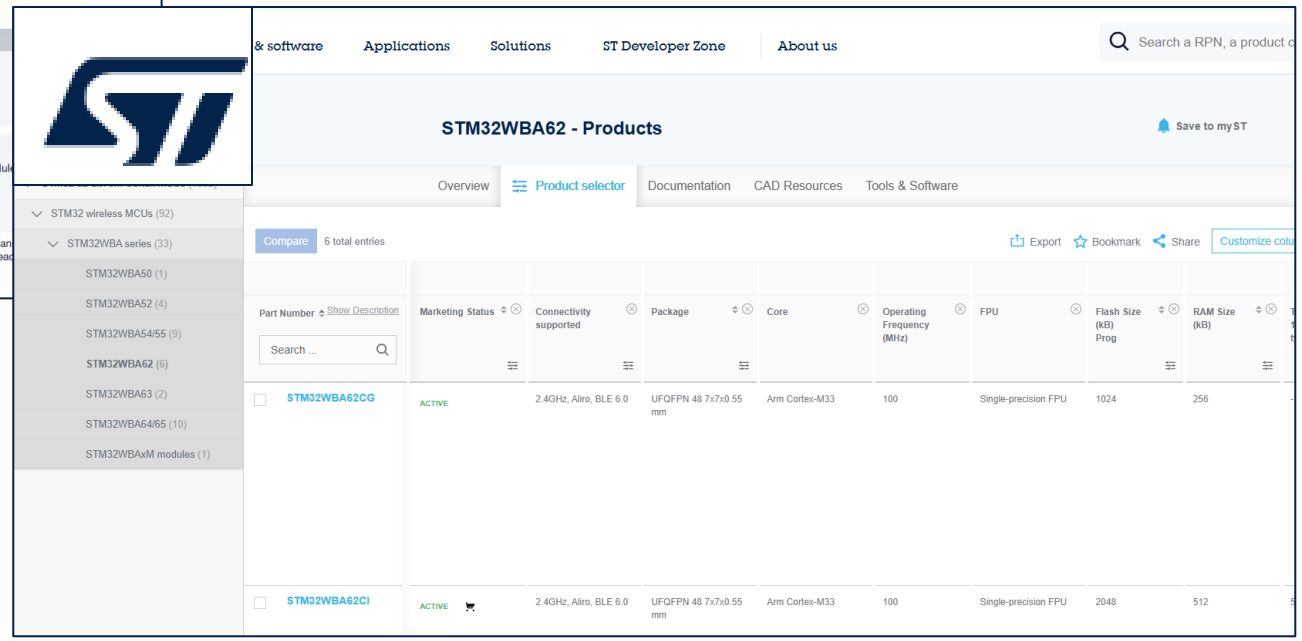
Let’s exchange, understand technical requirements and select the right part thanks to our product selector and our guidance. Let’s browse at st.com !

Select the right part (2/2)



From product overview

To product selector
package, features, BOM, GPIOS, …



Select the associated evaluation board

STM32WBA55CG is the right part

Discover our portfolio

Advanced features	STM32WBA65/64	STM32WBA55/54
	[1]Multi-protocol + audio Flash up to 2 Mbytes RAM up to 512 Kbytes SMPS/LDO	[2]Multi-protocol + audio Flash up to 1 Mbyte RAM up to 128 Kbytes SMPS/LDO

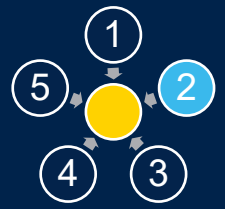
NUCLEO-WBA55CG as right vehicule for evaluation & POC

+

X-NUCLEO-IKS4A1
SHT40AD1B temp sensor

“Thanks for guidance, let’s move forward. How can you help to ease evaluation, and demonstrate this solution would meet my requirement ? ”

Software Package Selection & Ecosystem



ST Product Evaluation

Software Package Selection & Ecosystem
Choosing the Right Foundation

"How to start with ? There are so many sample codes and alternative "

From needs, and application scope, we will share guidance on way to start, and we can initiate a first POC

STM32 CubeMX

Let's start from scratch
Support to tune the NUCLEO
IOS mapping, BLE profile creation.

STM32 CubeWBA

Reuse ST code and derivate
support to tune as per as your needs
peripheral, enable ble features (ie : pairing, FOTA)

Legacy code Porting
support to port existing application
Move from existing project to STM32WBA NUCLEO

Team expertise on overall STM32 ecoystem on top of Wireless knowledge

Full Development Ecosystem
Beyond Packages - Our Daily Toolchain

"Thanks for starting point, now among the full set of IDEs , which one to use to build my POC ? "

Overall team expertise to support you on ST's environment development.

Choosing the Right Foundation
Outcome for our Wireless Sensor node POC

STM32 CubeWBA

NUCLEO-WBA55CG\Applications\BLE\BLE_HealthThermometer

STM32 CubeIDE

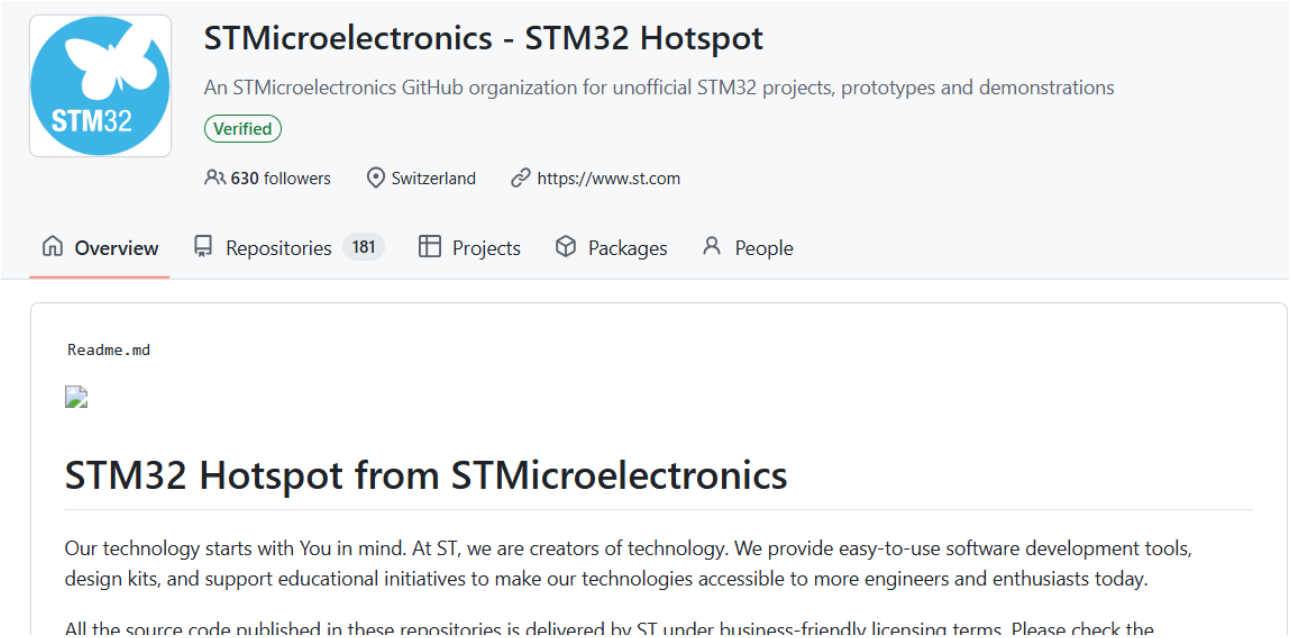
+

X-CUBE-MEMS1

integrate SHT40AD1B temperature sensor driver

POC material available at STM32 Hotspot

https://github.com/stm32-hotspot/ST-wireless-services-workshop



Software Package Selection & Ecosystem Choosing the Right Foundation

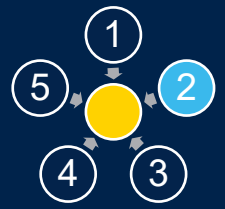
Not only limited to Bluetooth® Low Energy

STM32 CubeWBA

Samples code to address

Team expertise on overall 2.4Ghz Wireless technology ready to support you for your own needs to start your POC

PoC Development & validation



ST Product Evaluation

PoC Development & Porting Accelerate Your Start

Training on STM32 wireless solutions

- Wiki pages here to ease technology understanding
- MOOC at st community: https://community.st.com/t5/stm32-online-courses/stm32wba6-demo-workshop/ba-p/830444
- Wireless technology deep dive session on demand
- Hands-On to build wireless application (ie: BLE profile)

STM32WBA55 System architecture

- Browse SW package and understand system architecture
- Focus on specific part : flash management, driver, RF interrupt management

Senior expertise in embedded system & wireless communication

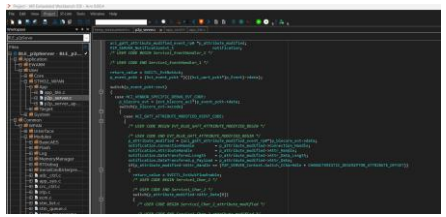
PoC Development & Porting From Concept to Running Code

"I am clear now on the setup to be used and product architecture.
How to collaborate to quickly enable my Bluetooth® Low Energy wireless sensor node POC ? "

https://ols.st.com/

remote

- "Let's Go and Code" sessions
- Real-time code sharing (GitHub, Teams)
- Step-by-step debugging guidance

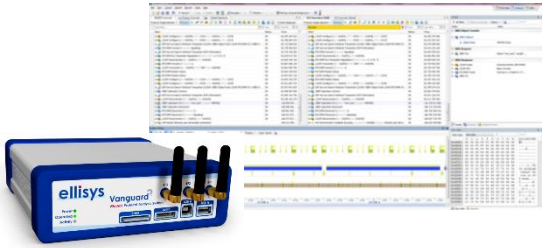


PoC Development & Porting
From Concept to Running Code

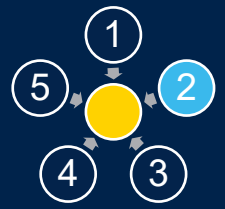
"I am facing challenges in my POC design, in system and wireless communication, could you help me ?"

On-site

- Critical system issue resolution
- Wireshark packet capture analysis
- Bluetooth® Low Energy air trace analysis and interpretation



PoC optimizations



ST Product Evaluation

Performance Metrics & Optimization

Beyond Functionality - Production-Ready Optimization

Expert Consulting

Partner with ST experts to tune your application for real-world reliability. We help you navigate complex design choices across **throughput requirements**, **architectural selection (NCP/RCP)**, and **secure OTA deployment strategies**.

Optimization Techniques We Apply

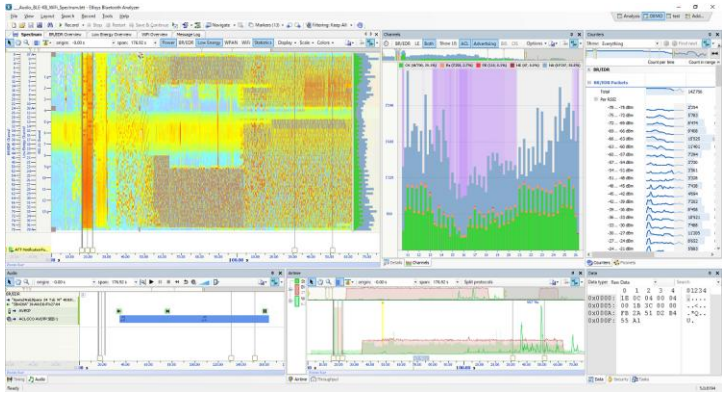
- RAM Retention configuration (save µA in Stop mode)
- I/O leakage reduction (pin configuration)
- RF profile tuning (TX power vs. range)
- Sleep mode transitions (STOP2 vs. Standby)
- Stack vs. heap optimization recommendations
- Flash/RAM usage breakdown & review

Performance validation

Tools set

"Which tools do you use to help me validate my PoC performance ? "

stm32cube monitor



Ellisys bluetooth analyzer

ST-LINK tools

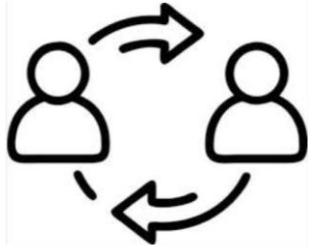
WIRESHARK

Tektronix®
Signal analyzer

We use best in class tools & SW to help you get the best of your application

Selection & Evaluation
Outcome and value proposal

Your success is our mission



- Comprehensive SW Packages : product-ready examples
- Deep technical competences: from system to wireless (protocols experts)
- Expertise access through a simple interface at https://ols.st.com/. This is us behind !
- Available training material: Wiki, MOOCS
  https://community.st.com/t5/stm32-online-courses/bg-p/STM32_learning
  https://community.st.com/t5/stm32-online-courses/stm32wba6-demo-workshop/ba-p/830444

Our technology starts with You

Find out more at www.st.com

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